Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 June 2026
Team ID	
Project Name	AI Powered Customer Ticket System
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture of the AI-Powered Customer Support Ticketing System is built entirely on the Salesforce Platform, following a cloud-native, multi-tenant architecture. The system is divided into three main zones — **User (Customer)**, **Salesforce Cloud (Core Platform)**, and **Admin/Support Team** — ensuring clear separation of interaction, processing, and administration layers.

The customer interacts with the system through the Salesforce-powered Customer Portal or Lightning App by logging in and submitting a support case. Once submitted, the case flows into the Salesforce Cloud, where it passes through Standard & Custom Objects (Account, Contact, Case, Support Category, SLA Policy) for structured storage. Validation Rules then ensure data accuracy and enforce business logic before the case proceeds further.

Record-Triggered Flows automate key backend processes such as case prioritization and SLA due-date calculation without manual intervention. The case data is then passed to Prompt Builder & Agentforce, Salesforce's generative AI components, which analyze the case details and generate an AI Suggested Reply for the support agent to review and use.

On the Admin/Support side, Support Agents and Managers access the processed case information through the Lightning Interface, where they can review AI-generated replies, update case status, and resolve customer issues. Finally, all case and performance data is aggregated into Reports & Dashboards, giving Support Managers real-time visibility into SLA compliance, case volume, and agent performance — enabling data-driven decision-making.

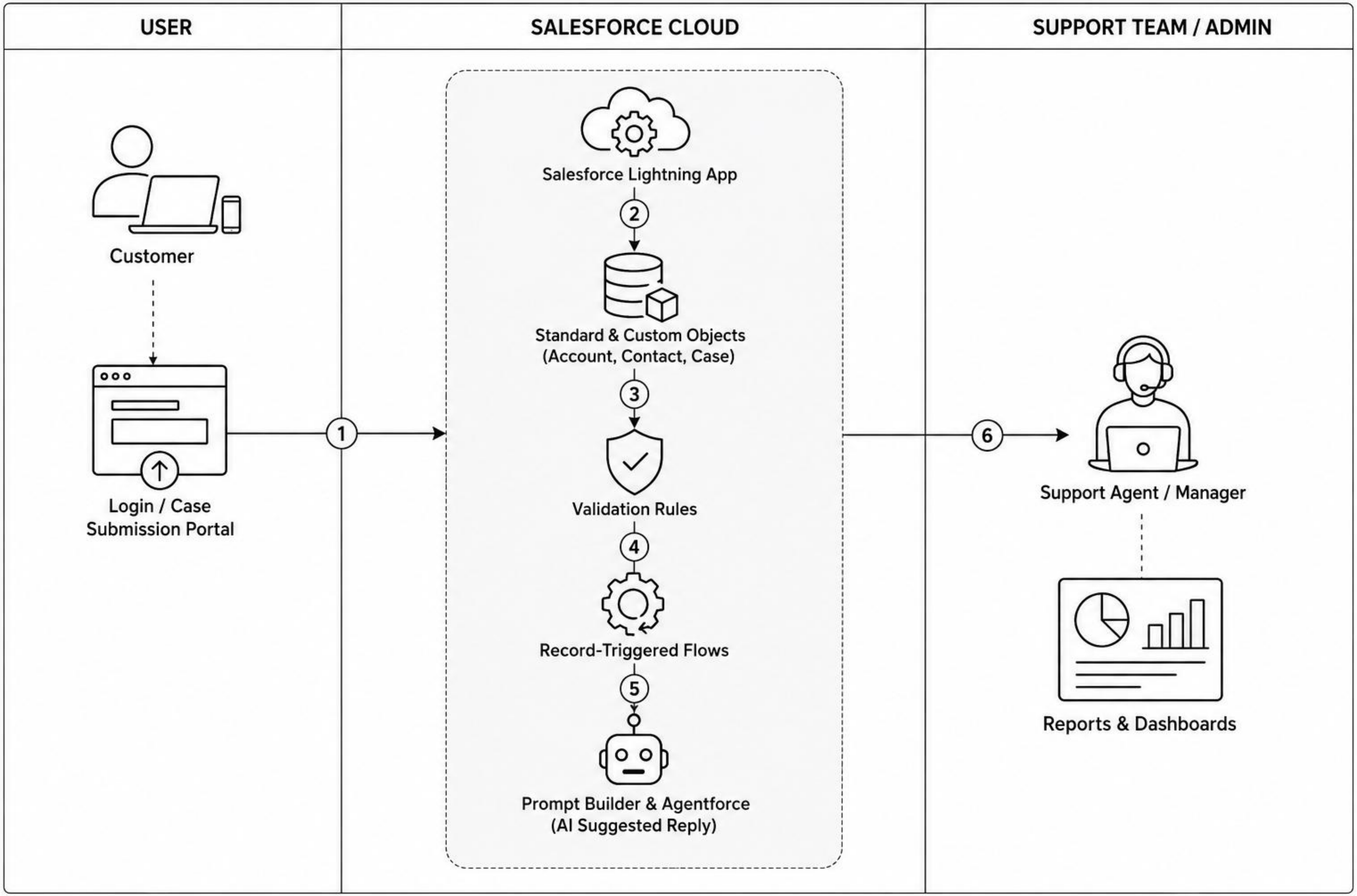
USER
SALESFORCE CLOUD
SUPPORT TEAM / ADMIN
Customer
Login / Case
Submission Portal
1
Salesforce Lightning App
2
Standard & Custom Objects
(Account, Contact, Case)
3
Validation Rules
4
Record-Triggered Flows
5
Prompt Builder & Agentforce
(AI Suggested Reply)
6
Support Agent / Manager
Reports & Dashboards

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How users interact with the application – Support Agent Console, Customer Portal, Manager Dashboard	Salesforce Lightning Web Components (LWC), Lightning App Builder
2.	Application Logic-1	Logic for Case creation, validation, and record management	Salesforce Apex, Validation Rules
3.	Application Logic-2	Logic for automated case classification, priority assignment, and SLA calculation	Record-Triggered Flows
4.	Application Logic-3	Logic for generating AI-based suggested replies for customer cases	Salesforce Prompt Builder & Agentforce
5.	Database	Data type, configuration, and object structure for storing records	Salesforce Standard & Custom Objects (Account, Contact, Case, Support Category, SLA Policy)
6.	Cloud Database	Database service on cloud	Salesforce Data Cloud (Salesforce Org Database)
7.	File Storage	File storage requirements for case attachments, documents, etc.	Salesforce Files / Salesforce Content Storage
8.	External API-1	Purpose of external API used in the application (e.g., email notifications)	Salesforce Email Services / SMTP API
9.	External API-2	Purpose of external API used for AI-generated reply suggestions	Agentforce / Einstein Generative AI API
10.	Machine Learning Model	Purpose of ML/AI model used	Einstein AI Model for Case Classification & Prompt Builder for AI Suggested Replies
11.	Infrastructure (Server / Cloud)	Application deployment on Local System / Cloud Local Server Configuration: N/A	Salesforce Cloud Platform (PaaS/SaaS)

		Cloud Server Configuration: Salesforce Multi-Tenant Cloud Infrastructure	

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Lightning Web Components (LWC) framework (built on open web standards - HTML, CSS, JavaScript)
2.	Security Implementations	Security/access controls implemented, use of firewalls, etc.	Role-Based Access Control, Profiles & Permission Sets, Object-Level & Field-Level Security, OAuth 2.0 Authentication, Salesforce Shield Encryption
3.	Scalable Architecture	Justify the scalability of architecture (3-tier, Microservices)	Multi-tenant Cloud Architecture with modular Objects, Flows, and AI components allowing independent scaling of Case volume and users
4.	Availability	Justify the availability of application (load balancers, distributed servers, etc.)	Salesforce Trust Platform with 99.9% uptime SLA, distributed data centers, automatic failover, and disaster recovery
5.	Performance	Design consideration for performance (requests/sec, caching, CDNs, etc.)	Salesforce Platform Cache, optimized SOQL queries, Lightning Component caching, and efficient Flow execution for fast report/dashboard generation